

POC-4: Secret: Mounting a Single Key with Custom Permissions

Steps:

1 . Reuse Secret db-credentials.

```
POC-3-SECRET-1 > ! secret.yaml > {} stringData > abc password
1  apiVersion: v1
2  kind: Secret
3  metadata:
4    name: db-credentials
5  type: Opaque
6  stringData:
7    username: admin
8    password: MyDBPassword123
```

2 . Create pod secret-subpath (toolbox image) that mounts only the password key to /etc/app/secret-password, using subPath, with defaultMode: 0400.

```
OC-4-SECRET2 > ! pod.yaml > {} spec > {} containers > {} 0 > {} volumeMounts > {} 0 > abc name
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: secret-subpath
5  spec:
6    containers:
7      - name: toolbox
8        image: docker.io/payalkharat/kubernetes-toolbox:latest
9        imagePullPolicy: IfNotPresent
10       volumeMounts:
11         - name: secret-volume
12           mountPath: /etc/app/secret-password
13           subPath: password
14           readOnly: true
15     volumes:
16       - name: secret-volume
17         secret:
18           secretName: db-credentials
19           defaultMode: 0400
20         items:
21           - key: password
22             path: password
```

3 . Verify: kubectl exec secret-subpath -- cat /etc/app/secret-password and kubectl exec secret-subpath -- ls -l /etc/app/secret-password.

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ ls
pod.yaml secret.yaml
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl apply -f secret.yaml
secret/db-credentials created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl get secrets
NAME      TYPE      DATA      AGE
db-credentials  Opaque    2          8s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl apply -f pod.yaml
pod/secret-subpath created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
secret-subpath  1/1     Running   0          5s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- cat /etc/app/se
cret-password
MyDBPassword123
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- ls -l /etc/app/
secret-password
-r----- 1 root root      15 Aug 20 07:21 /etc/app/secret-password
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- ls -la /etc/app
/
total 12
drwxr-xr-x 2 root root      4096 Aug 20 07:21 .
drwxr-xr-x 1 root root      4096 Aug 20 07:21 ..
-r----- 1 root root      15 Aug 20 07:21 secret-password
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ |

```

4 . **Expected result:** File contains the password value only; username is not present in the pod at all. File permission mode shows 0400 (read-only for owner), confirming defaultMode was applied.

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ ls
pod.yaml secret.yaml
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl apply -f secret.yaml
secret/db-credentials created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl get secrets
NAME      TYPE      DATA      AGE
db-credentials  Opaque    2          8s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl apply -f pod.yaml
pod/secret-subpath created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
secret-subpath  1/1     Running   0          5s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- cat /etc/app/se
cret-password
MyDBPassword123
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- ls -l /etc/app/
secret-password
-r----- 1 root root      15 Aug 20 07:21 /etc/app/secret-password
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ kubectl exec secret-subpath -- ls -la /etc/app
/
total 12
drwxr-xr-x 2 root root      4096 Aug 20 07:21 .
drwxr-xr-x 1 root root      4096 Aug 20 07:21 ..
-r----- 1 root root      15 Aug 20 07:21 secret-password
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-4-SECRET2$ |

```